AirCast

Estimating hourly PM2.5 with classical machine learning

Adrij Sarkar | B.Tech, second year | 21 October 2026

The research question

Can weather and calendar information estimate PM2.5 for a given hour, and where do the predictions fail? This study uses Beijing observations from 2010-2014 and evaluates models on later years. Same-hour weather is available to the model, so the experiment is contemporaneous estimation rather than a next-day forecast.

Executive findings

Selected model	Test RMSE	Test MAE	Test R^2
Random Forest	66.695 µg/m³	45.266 µg/m³	0.491

The forest was selected by 2013 validation RMSE. On 2014 data, test RMSE is about 6.4% lower than clipped Linear Regression. The model improves on the baseline but still misses many severe pollution episodes: 7.7% of test hours contribute 50.5% of squared error.

Study design at a glance

Stage	Period	Labelled hours
Training	2010-2012	24,418
Validation / model selection	2013	8,678
Final test	2014	8,661

Scope of this report. Numbers and charts are taken from the supplied executed notebook and matching result CSVs. Documentation was corrected and consistency checked; model fitting was not rerun during this report preparation. Original package versions were not recorded.

Reading guide

02 Data and EDA · 03 Preprocessing and model design · 04 Evaluation · 05 Error analysis · 06 Limits, improvements and sources

02 / Data and exploratory findings

UCI Beijing PM2.5 [1] contains 43,824 hourly observations (2010-2014). The target is measured at the US Embassy; weather is measured at Beijing Capital International Airport. Missing PM2.5 affects 2,067 observations (4.72%); no duplicate timestamps were found in the saved checks. Selected weather/calendar fields have no missing values.

Training PM2.5 has mean 97.82, median 73.0, standard deviation 89.25 and maximum 994 µg/m³. Positive skewness (1.70) shows a long upper tail. The mean therefore describes a different aspect of the distribution from the typical observed hour.

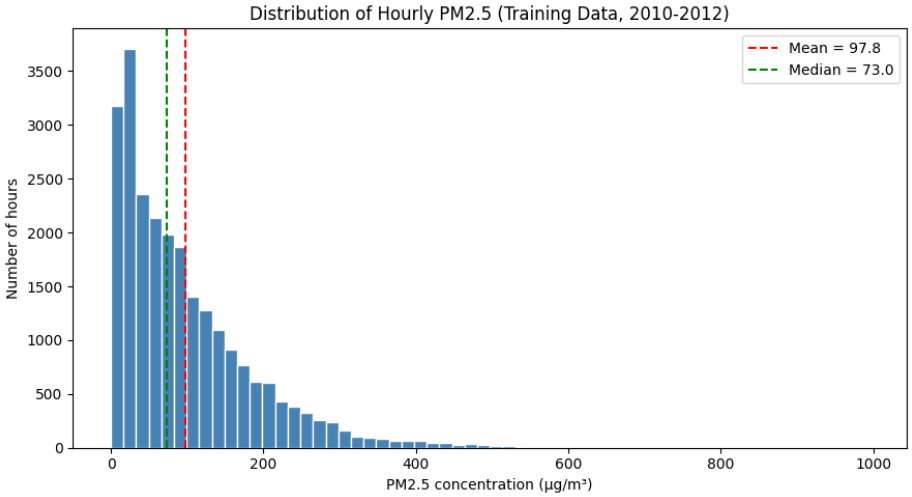


Figure 1. Training distribution. High readings pull the mean above the median.

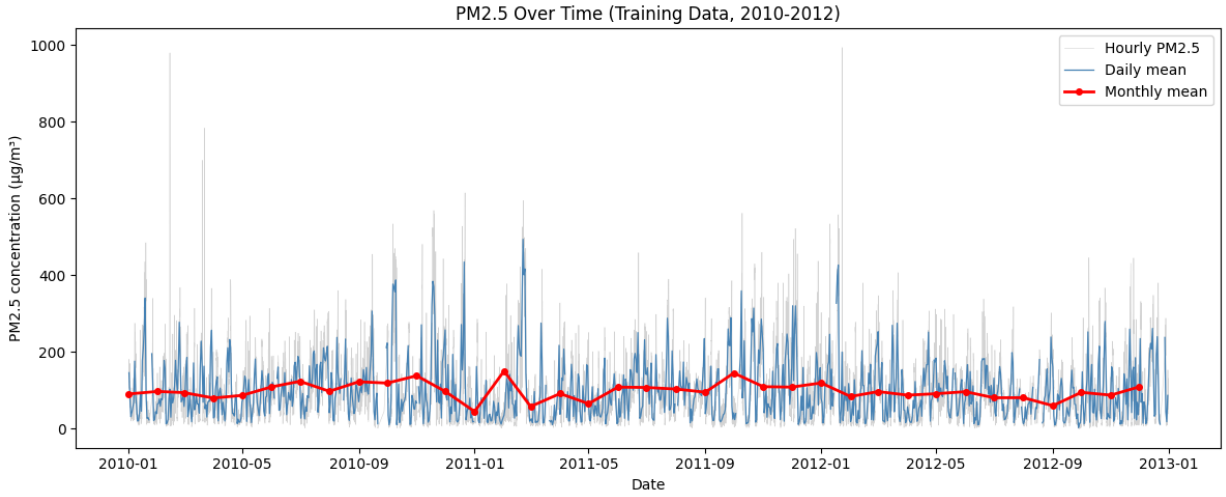


Figure 2. Hourly, daily and monthly PM2.5 during the training period.

Calendar-month averages are highest in October (118.2) and lowest in May (81.1). By hour, the maximum mean occurs at 01:00 (113.1); the minimum is at 15:00 (84.9), just below 16:00 (85.0). These are associations among measured hours, not proof of an atmospheric mechanism.

03 / Preprocessing and model design

Keep the evaluation chronologically separated

Rows without measured targets are excluded from supervised fitting and scoring; labels are not mean-filled. The full hourly table is retained for missingness analysis. Training-period missingness is 7.17% and varies by month, so conclusions may be affected by uneven observation coverage.

IQR flags 797 training PM2.5 hours above 305.5 µg/m³; |Z| > 3 flags 422 above approximately 365.6. These flags are not confirmed sensor errors. Plausible extremes are retained and cumulative wind is log-transformed rather than trimmed.

Feature engineering

Feature group	Construction / purpose
Relative humidity	Magnus approximation from TEMP and DEWP; capped at 100%.
Log cumulative wind	log(1 + Iws) compresses the strongly skewed right tail.
Precipitation indicator	1 when Is > 0 or Ir > 0; otherwise 0.
Hour cycle	sin(2π × hour / 24), cos(2π × hour / 24).
Month cycle	sin(2π × month / 12), cos(2π × month / 12).

Seven engineered columns contribute to 13 final inputs: 12 numeric plus wind direction. Raw year/day and any PM2.5 values are excluded from model inputs. The transformations above are row-wise and do not estimate parameters from future observations.

Training and selection

Numeric inputs use a training-fitted median imputer and standard scaler; wind direction is one-hot encoded. These steps are inside a Pipeline/ColumnTransformer. A mean baseline and Linear Regression are compared with Decision Tree depths 3, 5, 8, 12, 16 and unrestricted, plus Random Forest leaf sizes 5, 20 and 50.

The selected forest uses 200 trees, max_features=0.5, min_samples_leaf=20 and random_state=42. The selected tree has depth 8 and min_samples_leaf=20. Selection uses validation RMSE; final models remain fitted on 2010-2012 only. The three forest settings differ by less than 1 µg/m³ on one validation year.

04 / Evaluation and model comparison

All final metrics use predictions clipped at zero. Linear Regression improves from validation RMSE 73.987 before clipping to 73.430 after clipping; early notebook tables use raw values. The forest remains the selected model under either convention.

Model	Val RMSE	Test MAE	Test MSE	Test RMSE	Test R^2
Random Forest*	71.889	45.266	4448.241	66.695	0.491
Linear Reg. (clipped)	73.430	49.511	5075.071	71.240	0.420
Decision Tree	76.833	50.077	5395.474	73.454	0.383
Mean baseline	98.142	68.893	8747.145	93.526	-0.000

*Selected on validation RMSE. MAE/RMSE: µg/m³; MSE: (µg/m³)²; R^2: dimensionless. Values are the uploaded, rounded results.

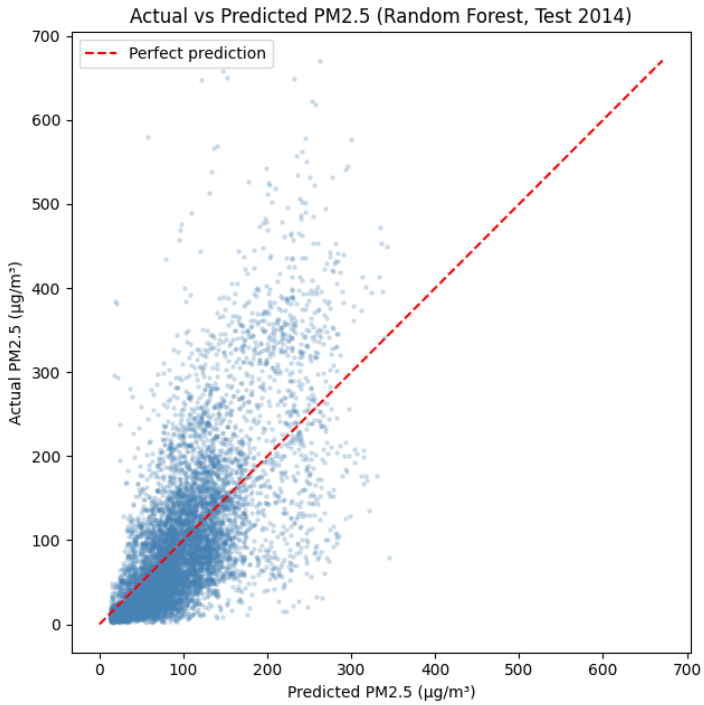


Figure 3. Perfect predictions would lie on the red diagonal; large observed concentrations are frequently underpredicted.

Interpretation. MAE measures average absolute deviation; RMSE gives greater weight to large misses. $R^2 = 1 - SSE/SST$, where SST is measured around the evaluated split's own mean. A value of 0.491 is not 49.1% accuracy. The fitted mean baseline uses the training mean and can therefore have a slightly negative test R^2.

Forest training R^2 is 0.653 versus validation 0.463 and test 0.491. The lower training error indicates a generalisation gap. A better test score than validation score does not establish less overfitting; the periods have different observation distributions.

05 / Errors and influential feature groups

Residual = actual - predicted; positive values mean underprediction. Overall mean residual is +1.61 µg/m³, but opposite errors cancel. Residual standard deviation rises from about 31 to 128 µg/m³ across predicted-concentration groups.

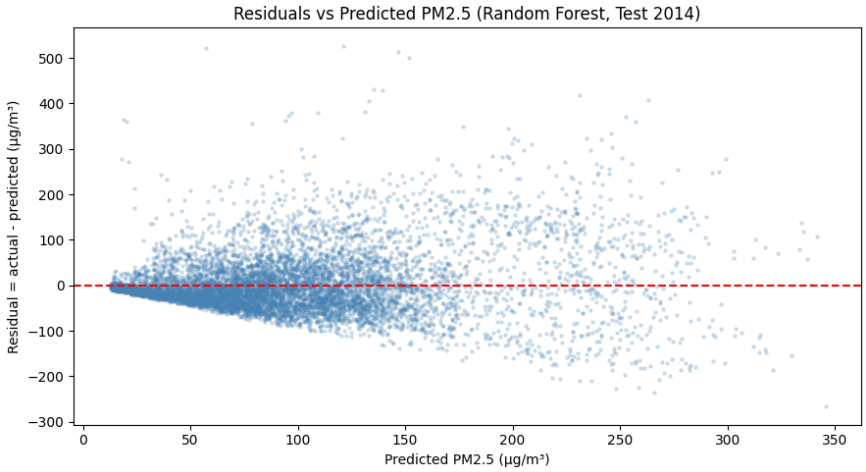


Figure 4. Residual spread increases at higher predictions, indicating uneven error variance.

Actual PM2.5 above 250 µg/m³ accounts for 7.7% of test hours and 50.5% of squared error. February has the largest monthly RMSE (113.4), while June is overpredicted on average (mean residual -36.7). The largest test prediction is 345.6 versus a largest actual value of 671.0; 345.6 is not a universal model cap.

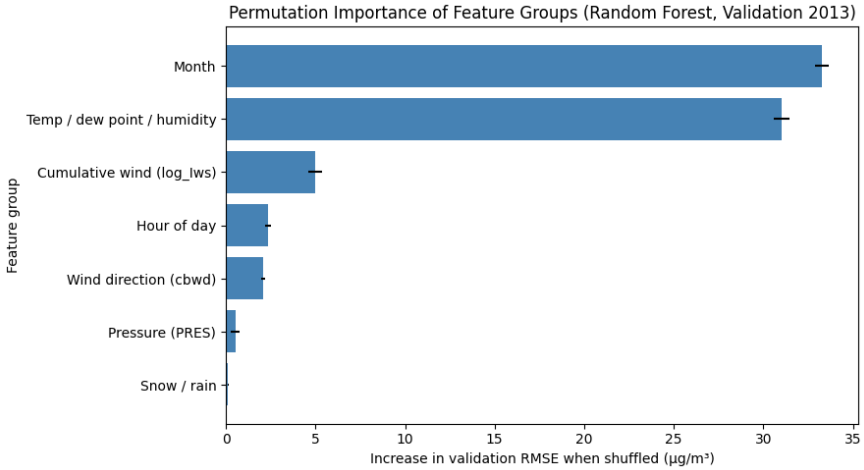


Figure 5. Validation permutation importance; bars show mean RMSE increase, with shuffle standard deviation over five repeats.

Correlated inputs make shuffling difficult to interpret. Refit ablations provide a second view: removing temperature/dew-point/humidity raises validation RMSE by 13.572; removing month raises it by 4.183; removing wind inputs raises it by 1.844. These effects measure predictive reliance under this experiment, not causation.

06 / Limitations, improvements and sources

What these results do not establish

Coverage. Missing labels are uneven over time and the weather sensor is not colocated with the pollution sensor. Results describe the measured observations and may not transfer to other cities or stations.

Inputs and severe events. Same-hour weather and calendar variables omit direct emissions and recent pollution history. Tree averaging can smooth rare extremes. A pollution-alert application would require further operational evaluation.

Evaluation. One validation year limits confidence in model ranking. Chronological splitting and training-only preprocessing reduce leakage risk but cannot independently prove the full development history. Since 2014 scores are now known, new model changes require a new untouched final evaluation period.

Concrete improvements

Use multiple expanding-window validation folds; compare features using the same splits and metric. Add historical weather trajectories on the complete hourly grid, respecting availability at prediction time. If lagged pollution is introduced, define a future forecast horizon and keep unavailable future weather out of the predictors. Inspect high-pollution errors separately from aggregate RMSE.

Reproducibility and assistance

The repository contains the corrected notebook, three supplied result CSVs, eight recovered PNG plots and JPG copies, dependency list, methodology, reflections and reproduction notes. Original versions and a raw data snapshot were not supplied; the notebook downloads UCI data. A clean-kernel rerun remains to be confirmed.

Gemini assistance with pipeline/diagnostic design and formatting was disclosed in the original notebook. ChatGPT assisted with coding discussion, interpretation, saved-output review, corrections and documentation. The author remains responsible for understanding and verifying the submitted work.

References

[1] Chen, S. (2015). Beijing PM2.5. UCI Machine Learning Repository. DOI: 10.24432/C5JS49. CC BY 4.0.

[2] Liang et al. (2015). Assessing Beijing’s PM2.5 pollution: severity, weather impact, APEC and winter heating. Proc. R. Soc. A, 471, 20150257. DOI: 10.1098/rspa.2015.0257.

[3] Alduchov & Eskridge (1996). Improved Magnus Form Approximation of Saturation Vapor Pressure. Journal of Applied Meteorology, 35, 601-609.

[4] Scikit-learn official documentation: RandomForestRegressor, common pitfalls and R^2 score. Full clickable references are provided in docs/SOURCES.md.

[5] Numerical evidence: supplied AirCast notebook and model_comparison, model_comparison_final and ablation_validation CSVs.